

Ramdeobaba University , Nagpur

Department Of Mathematics

Program : B.Tech.(EE) First Year Semester I, Session 2026-27

Course Name : Engineering Mathematics I

Course Code : 26HS03TP0106

Expirement NO.3 : 3D Plotting Techniques for ODE-Related Engineering Models

Name Of Student : Khushmit Nitin Wanjari

Section : J-1 B-2

Roll No. : 31

Date : 24/08/2026

3D plots

1.Arrow can be drawn in 3d by mentioning starting point and end point.

In [28]: `arrow3d((1,1,1),(2,2,2),4,color="purple")`

Out[28]: 

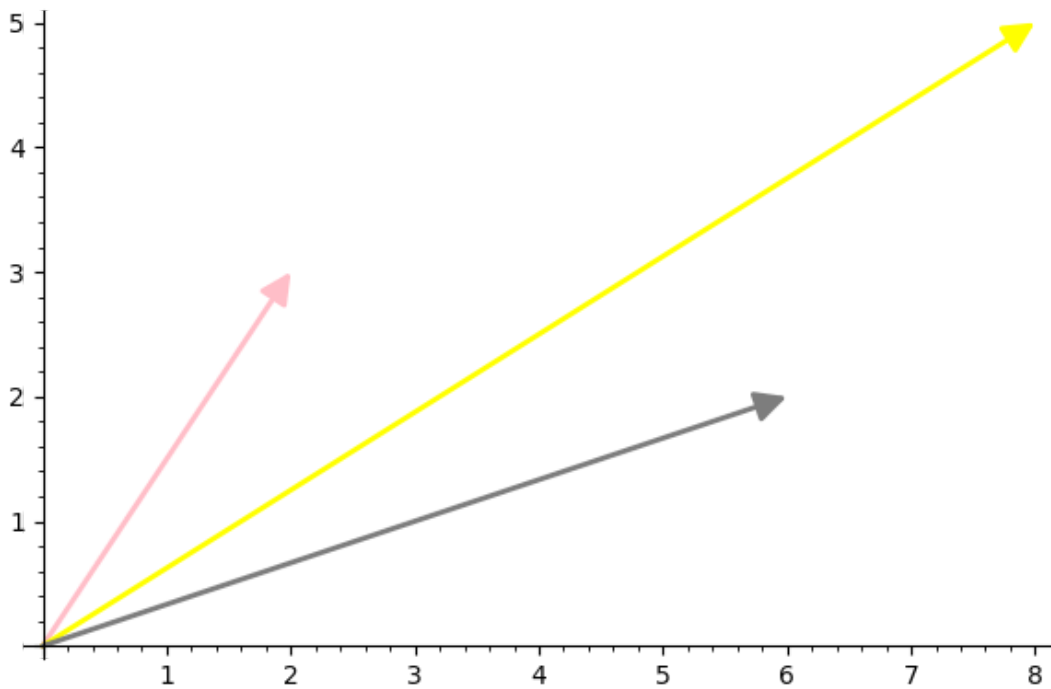


In [31]: `a=arrow3d((0,0,0),(1,1,1),6)
b=arrow3d((0,0,0),(1,2,3),4,color='blue')
c=arrow3d((0,0,0),(1,-2,1),3,color='pink')
show(a+b+c)`



2.vectors can be drawn if we know it's x,y,z components.

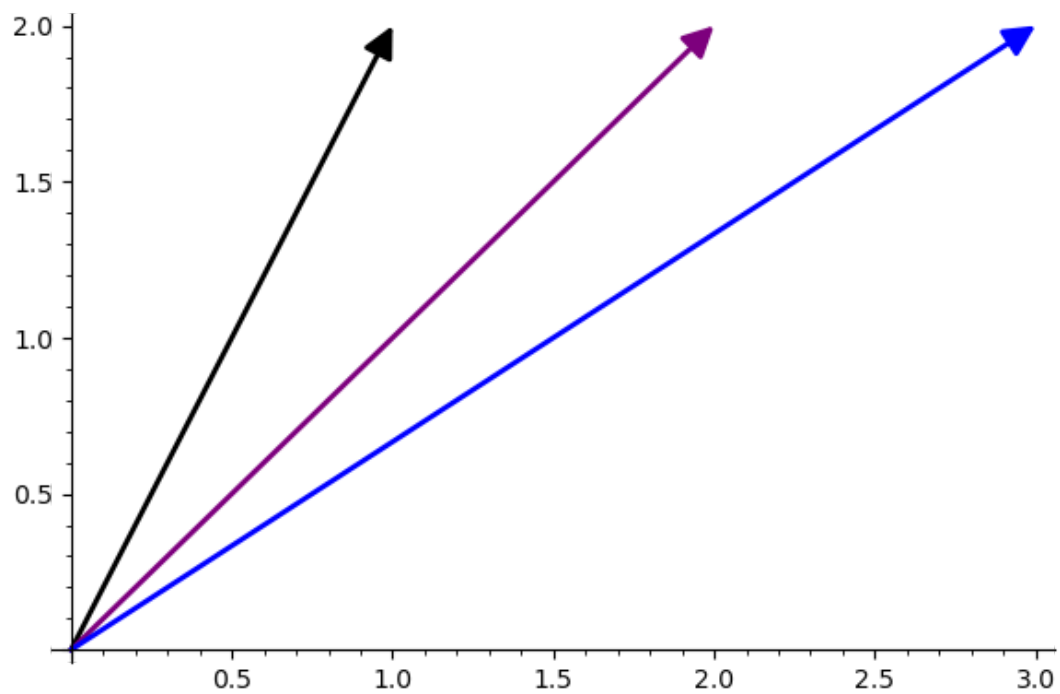
```
In [9]: a=vector([2,3])  
b=vector([6,2])  
c=a+b  
p1=plot(a,color="pink")  
p2=plot(c,color="yellow")  
p3=plot(b,color="grey")  
show(p1+p2+p3)
```



```
In [34]: a=vector([1,2,3])
b=vector([2,2,-1])
c=vector([3,2,-4])
p1=plot(a,color="red")
p2=plot(b,color='orange')
p3=plot(c)
show(p1+p2+p3)
```



```
In [13]: a=vector([1,2])  
b=vector([2,2])  
c=vector([3,2])  
p1=plot(a,color='black')  
p2=plot(b,color='purple')  
p3=plot(c)  
show(p1+p2+p3)
```



3D plots

1.plot $x^n + y^n + z^n - 4 = 0$ by using implicit plot 3d command for different values of n .

```
In [35]: var('x,y,z')
n=2
f(x,y,z)=x^n+y^n+z^n-4
implicit_plot3d(f,(x,-2,2),(y,-2,2),(z,-2,2),opacity=.5)
```

Out[35]: 

```
In [36]: var('x,y,z')
n=3
f(x,y,z)=x^n+y^n+z^n-4
implicit_plot3d(f,(x,-2,2),(y,-2,2),(z,-2,2), color='darkorange')
```

Out[36]: 



2. plot $x^2+y^2-2z=0$ with $(x, -2,2),(y,-2,2),(z,0,2)$

```
In [37]: implicit_plot3d(x^2+y^2-2*z,(x,-2,2),(y,-2,2),(z,0,2),color='green')
```

Out[37]: 



1. Combine graph example

```
In [32]: p1=implicit_plot3d(x^2+y^2-4,(x,-2,2),(y,-2,2),(z,0,2),color='red')
p2=implicit_plot3d(x^2+y^2-2*z,(x,-2,2),(y,-2,2),(z,0,2),color='blue')
show(p1+p2)
```



Surface Of Revolution

1. surface of revolution by revolving $f(x) = x^3$ about z-axis through the angle to 0 2π

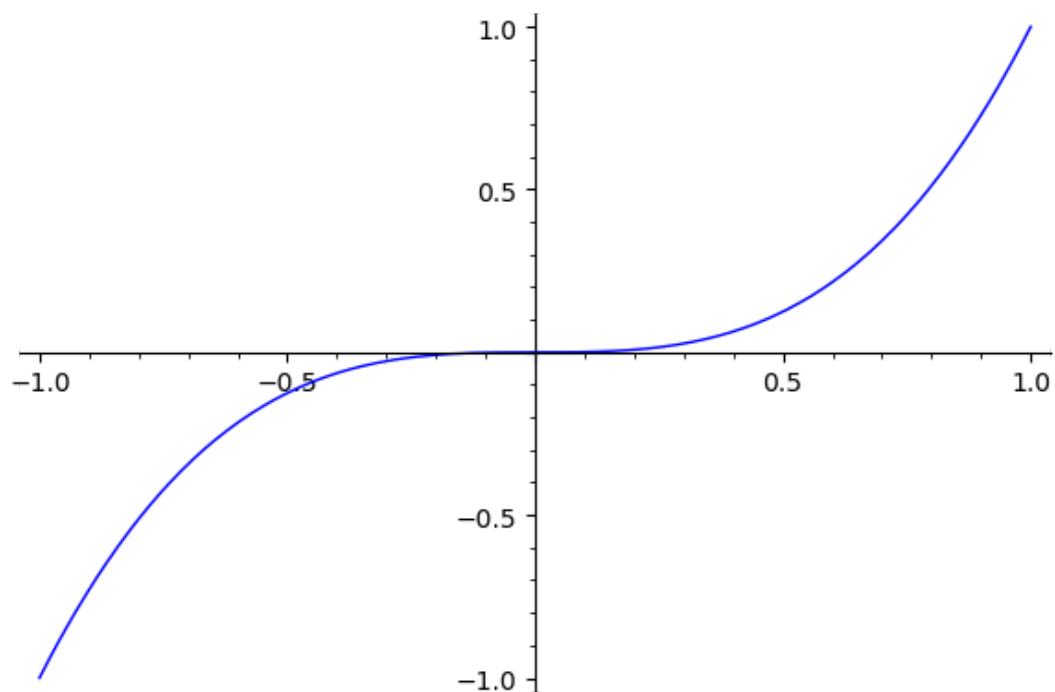
```
In [33]: revolution_plot3d(x^3,(x,-2,2),parallel_axis='z',show_curve=True,color="yellow")
```

```
Out[33]: 
```

2.Surface of revolution of curve $f(x) = x^3$ about x-axis through angle to 0 2π .

In [26]: `plot(x^3)`

Out[26]:



In [38]: `revolution_plot3d(x^3,(x,-2,2),parallel_axis='y',show_curve=True,color="red")`

Out[38]:



```
In [39]: revolution_plot3d(x^3,(x,-2,2),parallel_axis='z',show_curve=True,color="lightgreen")
```

Out[39]: 



3.Surface of revolution of curve $f(x) = x^3$ about x-axis through the angle to 0π .

```
In [40]: revolution_plot3d(x^3,(x,-2,2),(0,2*pi),parallel_axis='x',show_curve=True,color='pink')
```

Out[40]: 



4. Surface of revolution of curve $f(x) = x^2$ about x-axis through the angle to $0\ 2\pi$.

```
In [41]: revolution_plot3d(x^2,(x,0,2),(0,2*pi),parallel_axis='y',show_curve=True)
```

```
Out[41]: 
```

5.Surface of revolution of parametric curve $x = \cos t$, $y = \sin t$.

```
In [42]: var('t')  
         revolution_plot3d((cos(t),sin(t)),(t,0,pi),parallel_axis='z',color= 'grey')
```

Out[42]: 

6.Surface of revolution of curve $f(x) = x$ when x is from 0 to 4 about x axis.

```
In [43]: revolution_plot3d(x,(x,0,4),parallel_axis='x',color='violet',figsize="0.5")
```

```
Out[43]: 
```

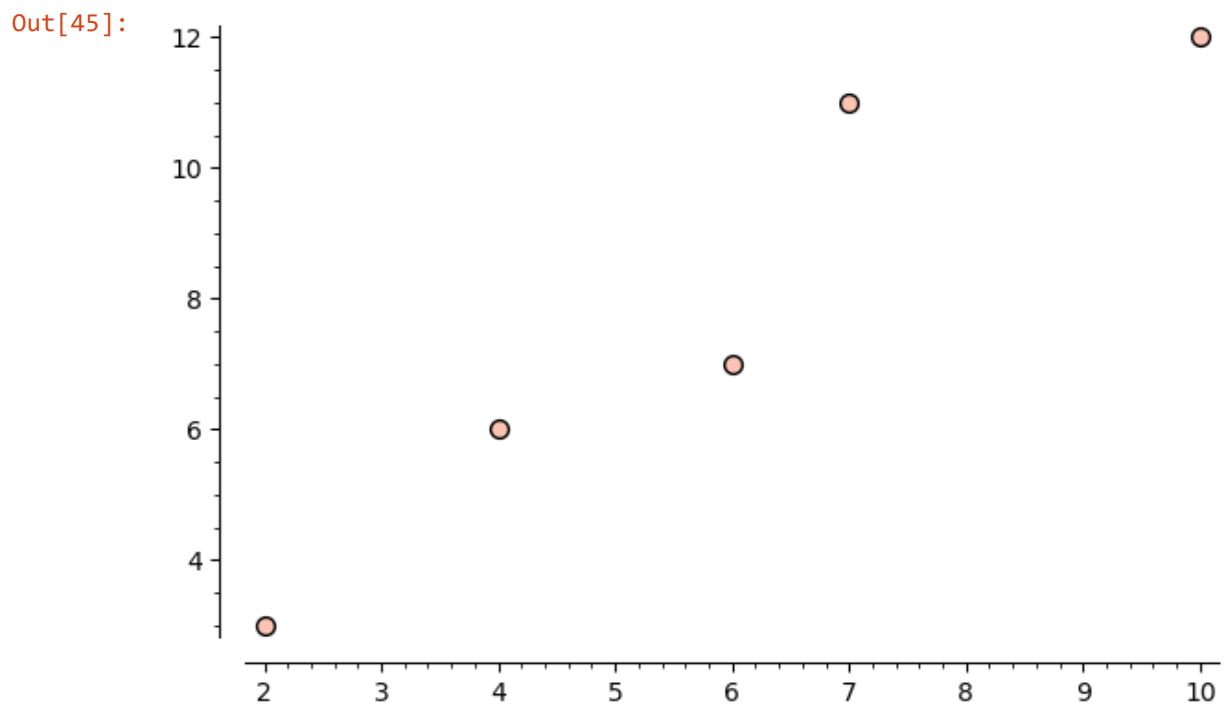
7.surface of revolution of curve $f(x) = x$ when x is from -4 to 4 about x axis.

```
In [44]: revolution_plot3d(x,(x,-4,4),(0,2*pi),parallel_axis='x',show_curve=True,color="red")
```

Out[44]: 

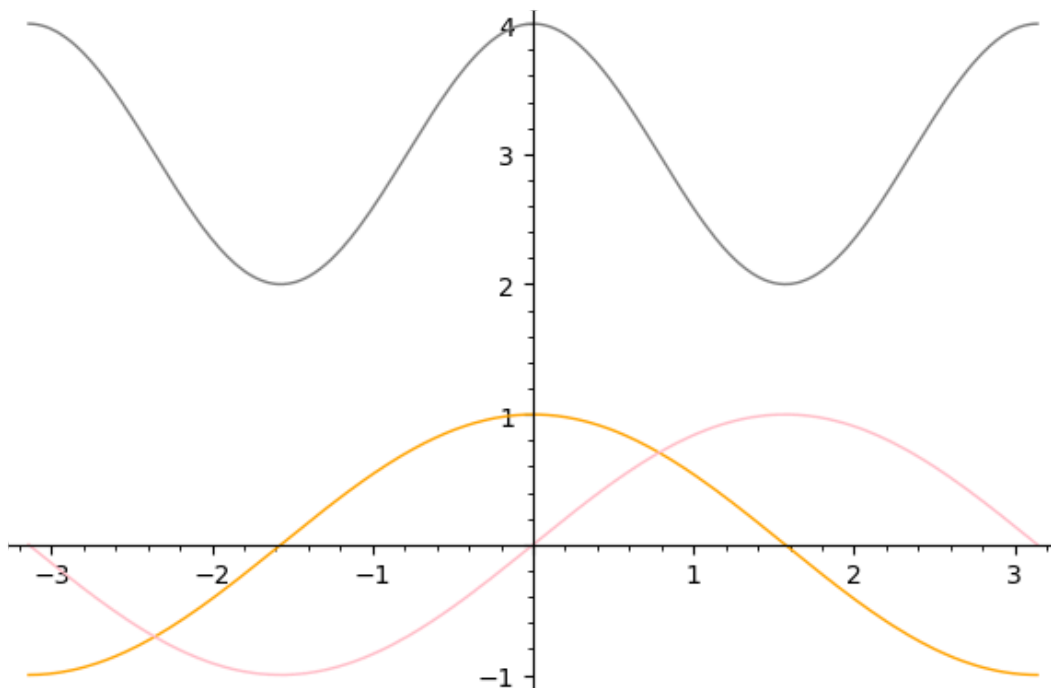
i

```
In [45]: a=[(2,3),(4,6),(6,7),(7,11),(10,12)]
scatter_plot(a)
```



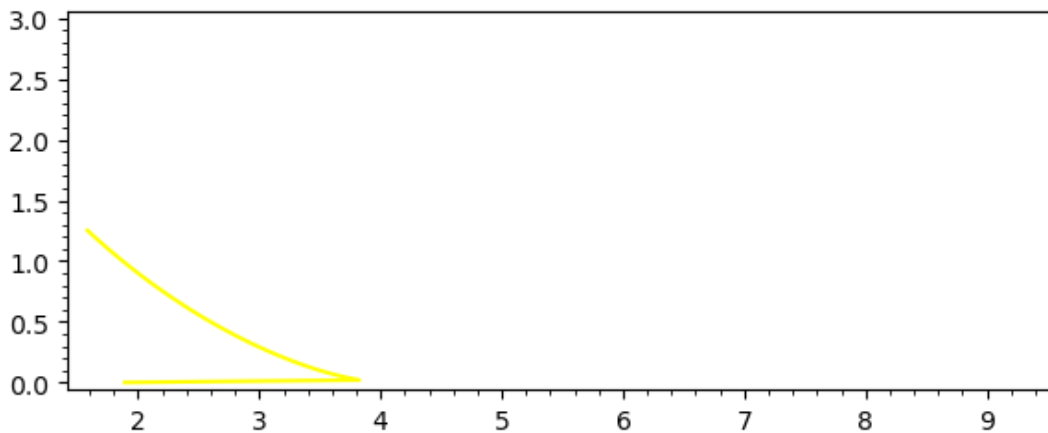
```
In [46]: plot(cos(x),(-pi,pi),color='orange')+plot(sin(x),(-pi,pi),color='pink')+plot(3 +cos(2*x),(-pi,pi),color='grey')
```

Out[46]:

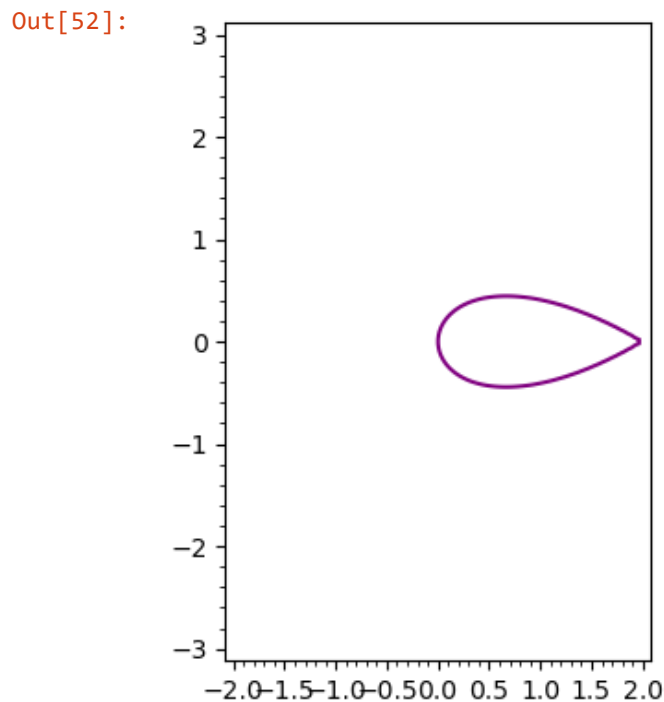


```
In [47]: var('x,y')
n=2/3
f(x,y)=x^n+y^n-4^n
implicit_plot(f,(x,pi/2,3*pi),(y,0,3),color='yellow')
```

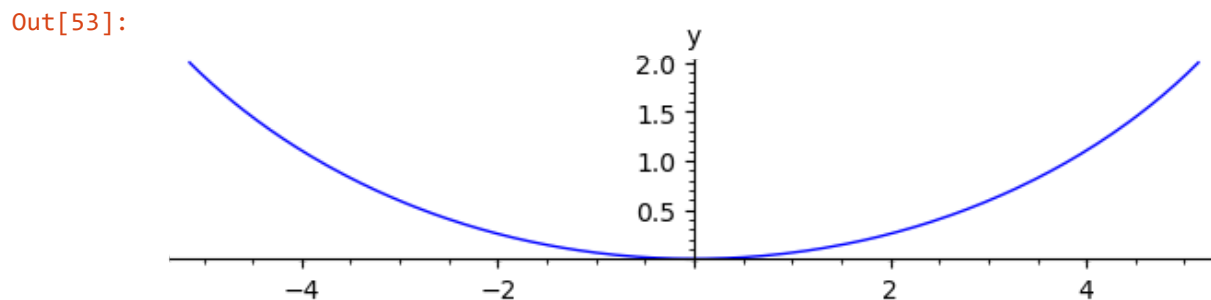
Out[47]:




```
In [52]: var('x,y')
f(x,y)=6*y^2-x*(x-2)^2
implicit_plot(f,(x,-2,2),(y,-3,3),color='purple')
```



```
In [53]: var('t')
parametric_plot([2*(t+sin(t)),2*(1-cos(t))],(t,-pi/2,pi/2),title='y')
```



Exercises

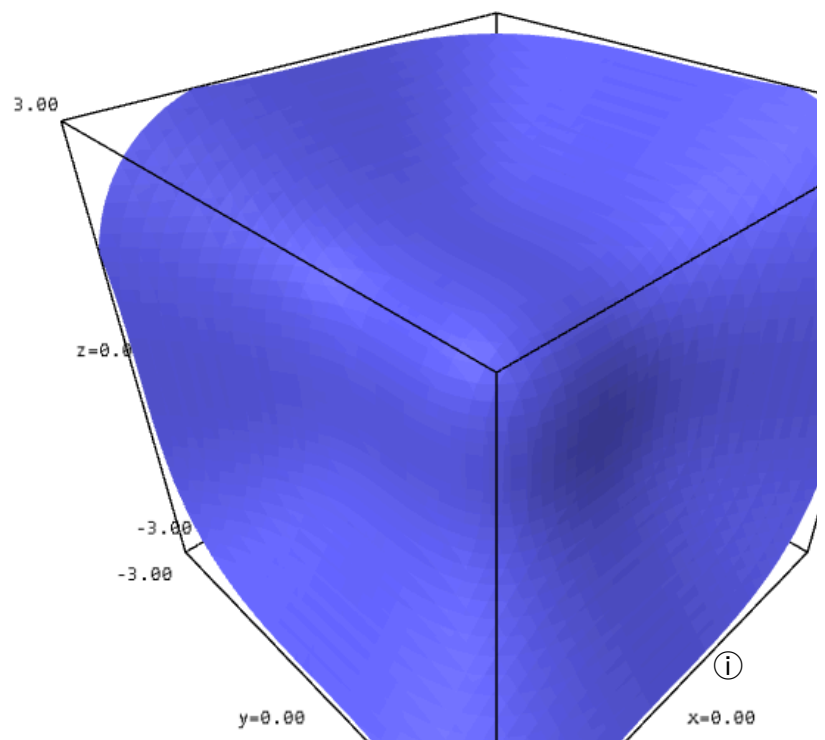
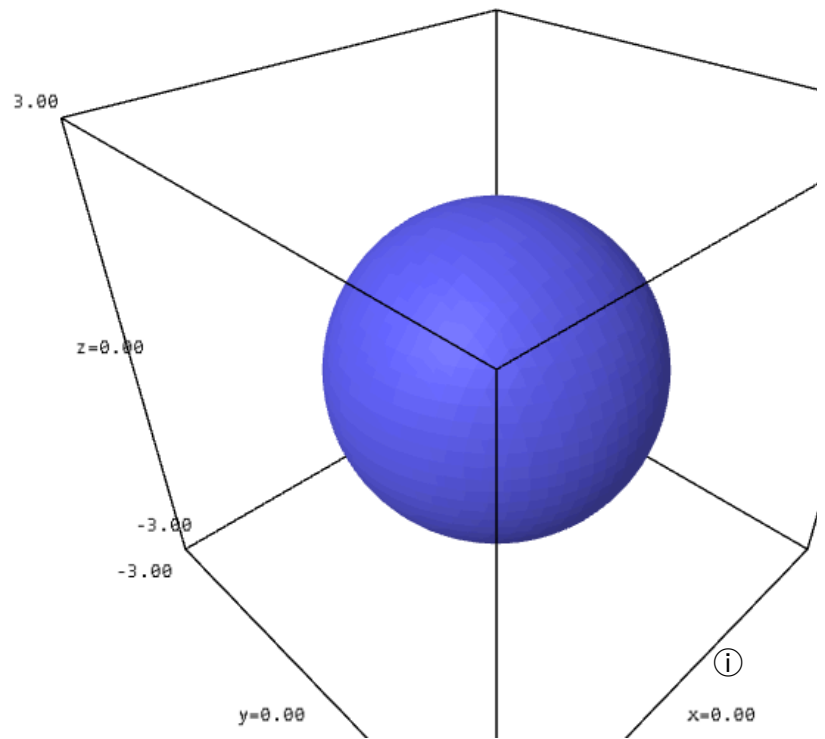
1) Find the graph of $f(x, y, z) = x^n + y^n + z^n - 4$ and analyze the graph by considering odd and even values of n .

```
In [20]: var('x y z')
f(x, y, z, n) = x^n + y^n + z^n - 4

p1 = implicit_plot3d(f(x, y, z, 2) == 0, (x, -3, 3), (y, -3, 3), (z, -3, 3))

p2 = implicit_plot3d(f(x, y, z, 3) == 0, (x, -3, 3), (y, -3, 3), (z, -3, 3))
```

```
show(p1)  
show(p2)
```



2) Find the surface of revolution of $y=x^3$ around z axis through an angle 0 to 2π and through x- axis through an angle 0 to 2π and 0 to π .

```
In [21]: var('u v')

surf_z = revolution_plot3d(u^3, (u, 0, 2), (v, 0, 2*pi), axis=(0, 0),color="red")

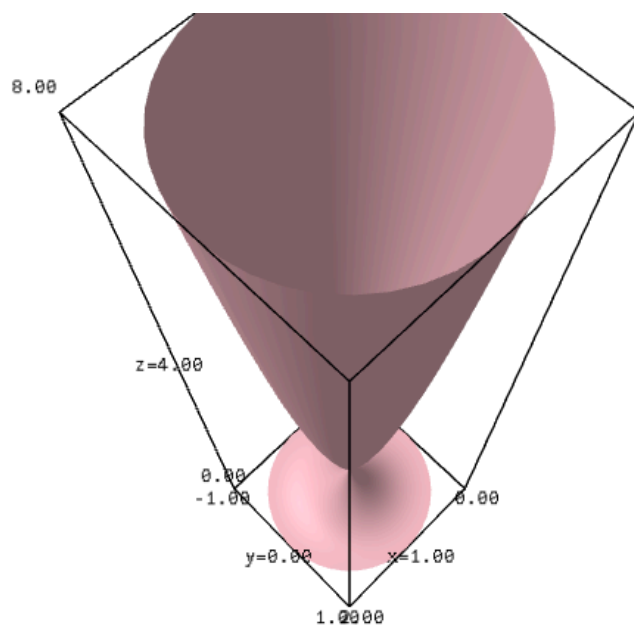
surf_x1 = revolution_plot3d(u^3, (u, 0, 2), (v, 0, 2*pi), axis=(1, 0),color="pink")

surf_x2 = revolution_plot3d(u^3, (u, 0, 2), (v, 0, pi), axis=(1, 0),color="royalblue")

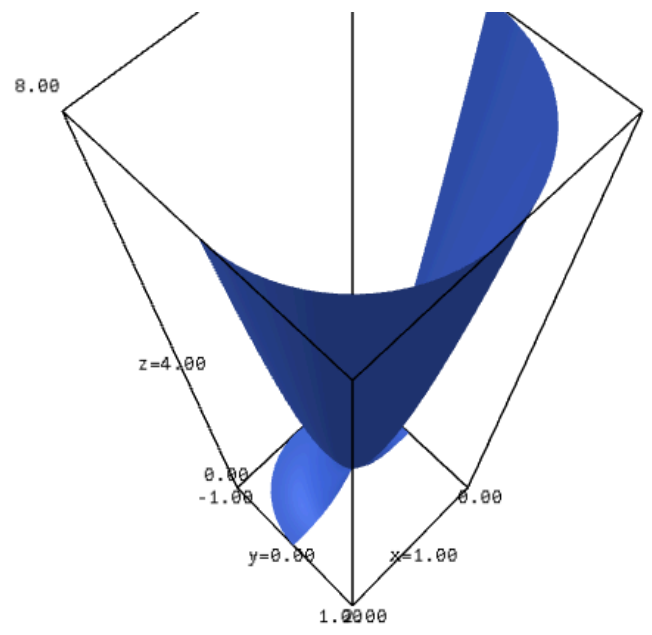
show(surf_z)
show(surf_x1)
show(surf_x2)
```



i



i



i

3) Find the surface of revolution of $y=x^3+x$ around z axis through an angle 0 to π and through x-axis through an angle 0 to 2π and 0 to 2π .

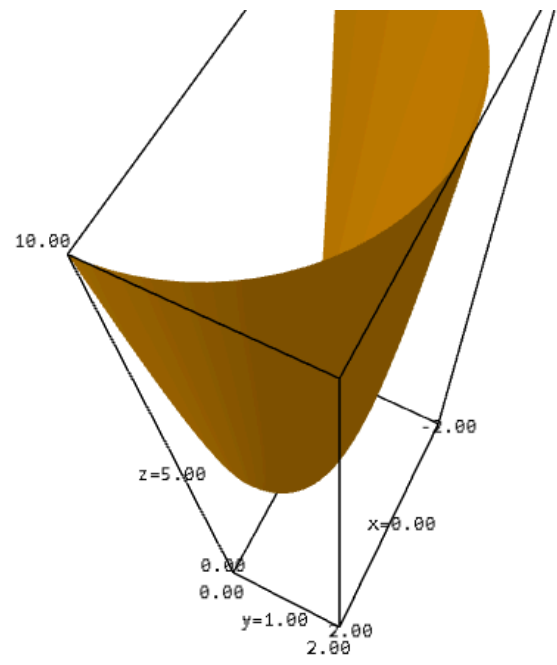
```
In [22]: var('a b')

p_z = revolution_plot3d(a^3 + a, (a, 0, 2), (b, 0, pi), axis=(0, 0),color="orange")

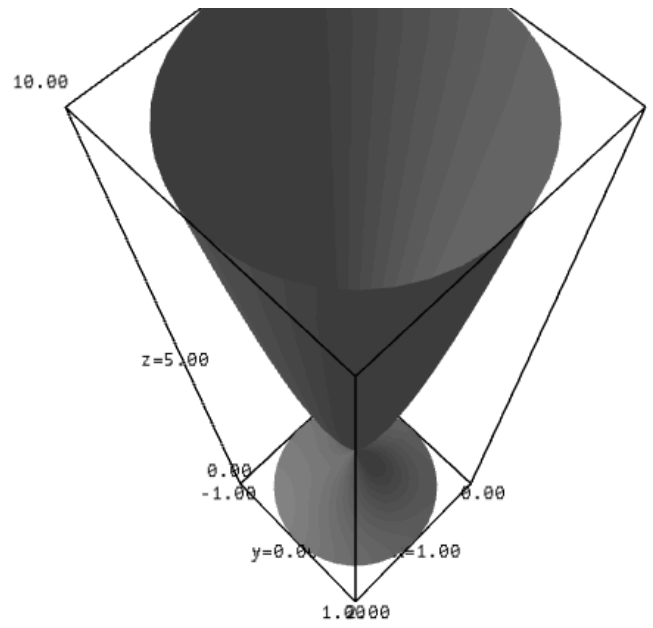
p_x1 = revolution_plot3d(a^3 + a, (a, 0, 2), (b, 0, 2*pi), axis=(1, 0),color="green")

p_x2 = revolution_plot3d(a^3 + a, (a, 0, 2), (b, 0, 2*pi), axis=(1, 0),color="violet")

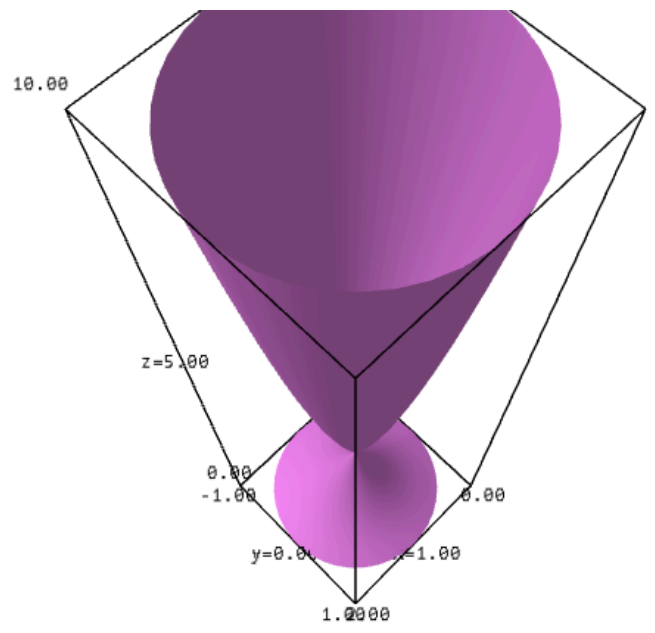
show(p_z)
show(p_x1)
show(p_x2)
```



i



i



i

4) Plot the graph of the following curves:

(a) $\sin(x)\cos(y) + \sin(y)\cos(z) + \sin(z)\cos(x) = 0$ within the range $x \in [-\pi, \pi]$, $y \in [-\pi, \pi]$, $z \in [-\pi, \pi]$.

(b) $x^3 - 3xy^2 - z = 0$ within the range $x \in [-2, 2]$, $y \in [-2, 2]$, $z \in [-2, 2]$.

(c) $(R - \sqrt{x^2 + y^2})^2 + z^2 - r^2 = 0$, $R > r$ within the range $x \in [-3, 3]$, $y \in [-3, 3]$, $z \in [-2, 2]$ for different values of R, r .

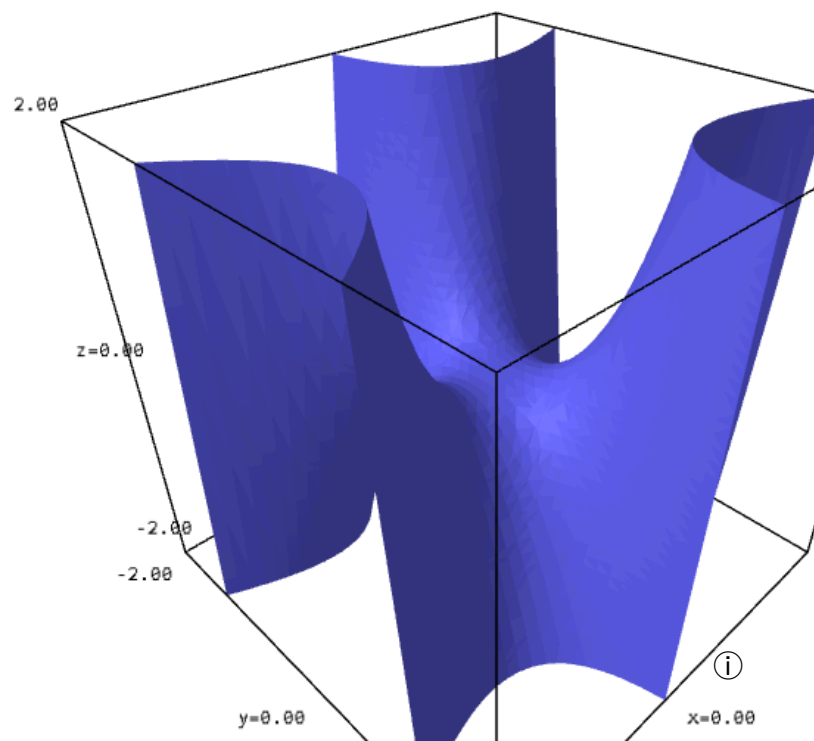
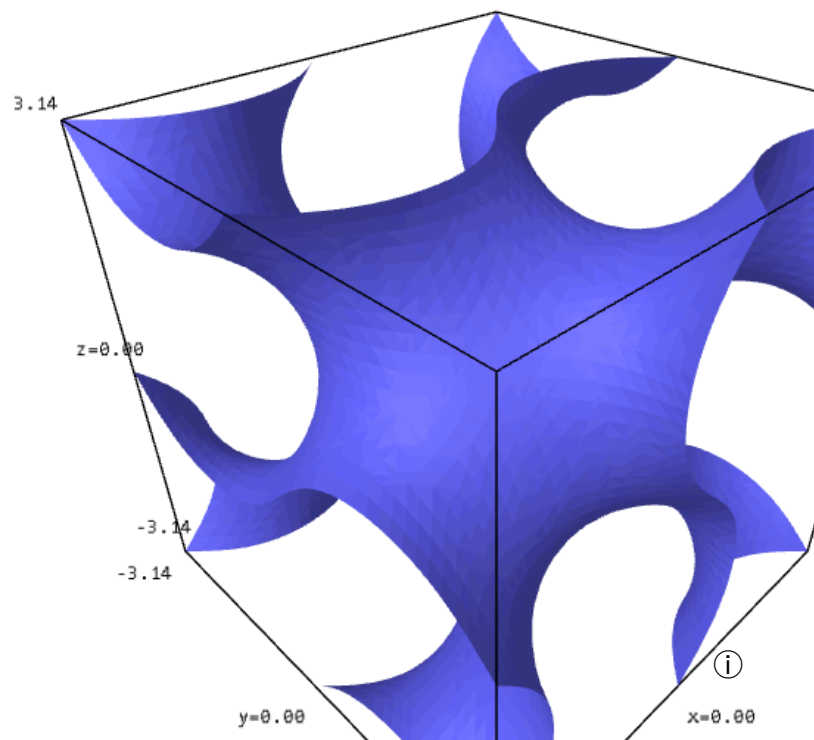
```
In [29]: var('x y z')

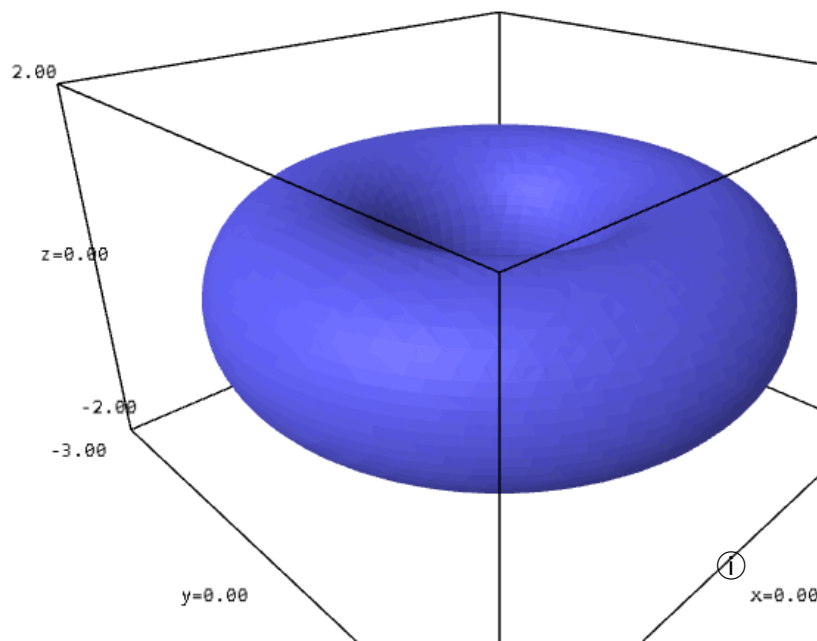
p4a = implicit_plot3d(
    sin(x) * cos(y) + sin(y) * cos(z) + sin(z) * cos(x) == 0,
    (x, -pi, pi),
    (y, -pi, pi),
    (z, -pi, pi),
)

p4b = implicit_plot3d(x^3 - 3 * x * y^2 - z == 0, (x, -2, 2), (y, -2, 2), (z, -2, 2))

R, r = 2, 1
p4c = implicit_plot3d(
    (R - sqrt(x^2 + y^2))^2 + z^2 - r^2 == 0, (x, -3, 3), (y, -3, 3), (z, -2, 2)
)

show(p4a)
show(p4b)
show(p4c)
```





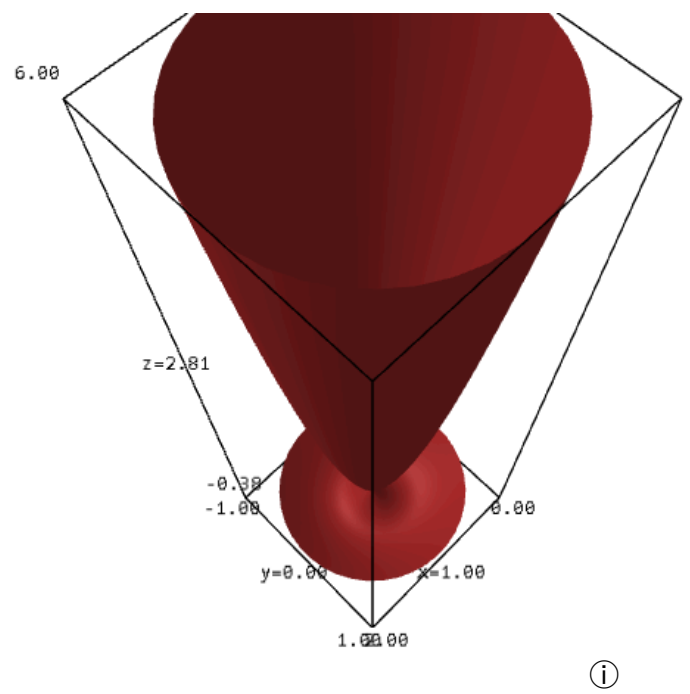
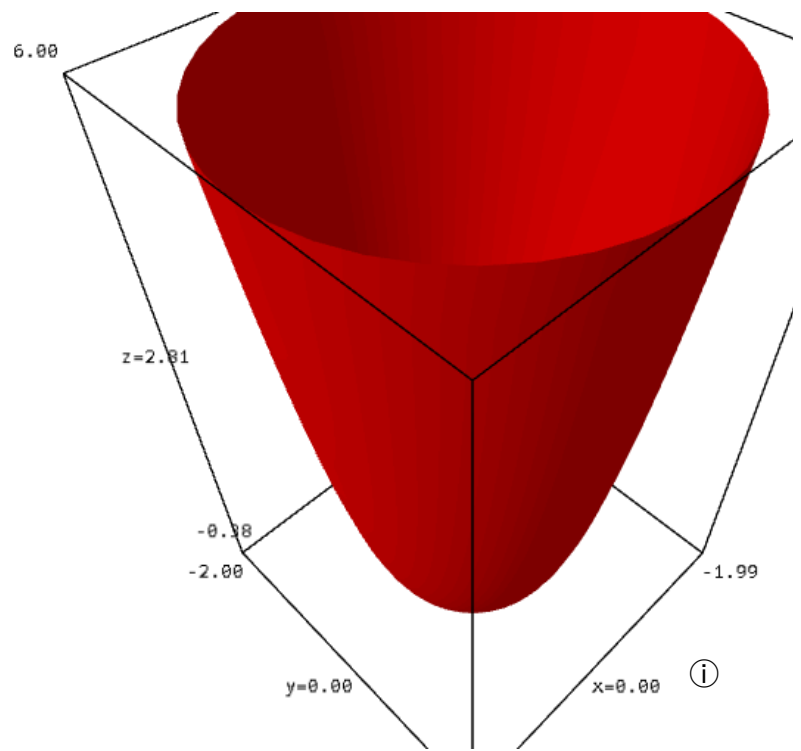
5) Find the revolution plot along x, y, z axes of the following curves with 360 degree rotation:

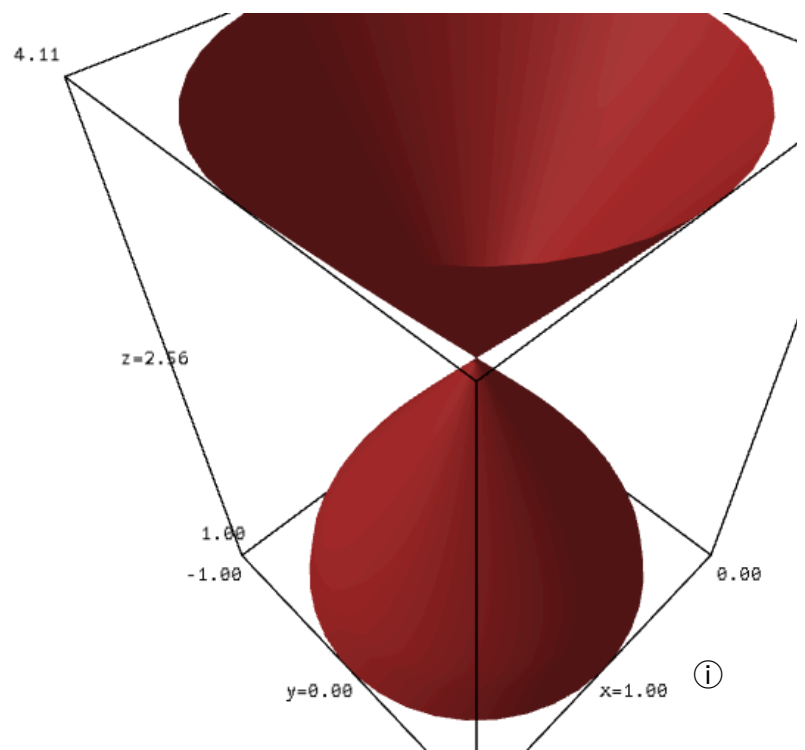
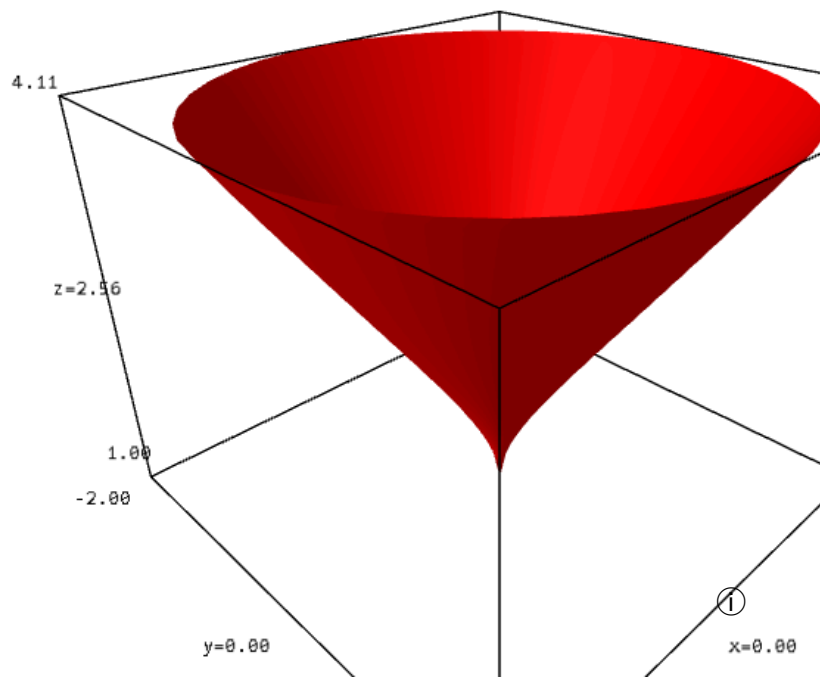
(a) $y = x^3 - x$

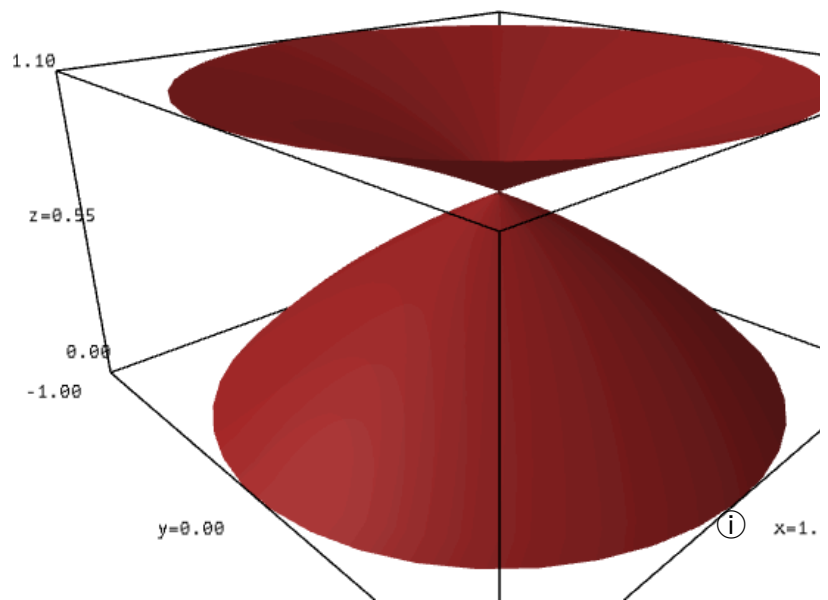
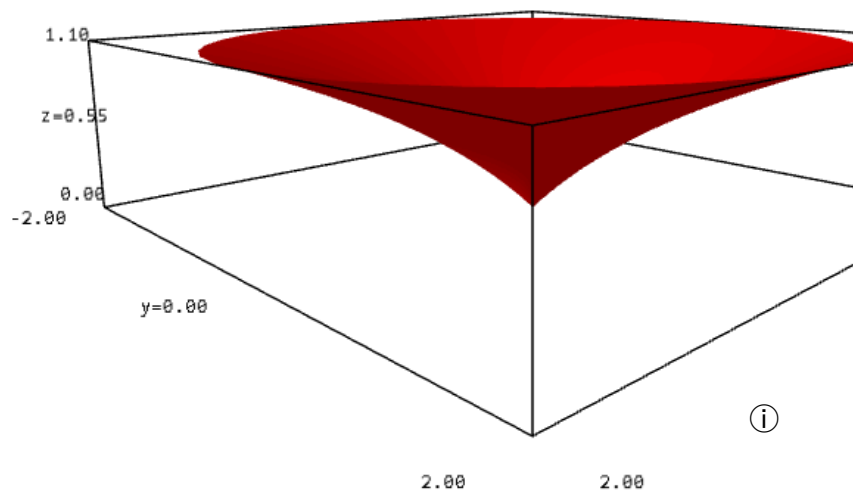
(b) $y = e^{\sqrt{x}}$

(c) $y = \ln(x + 1)$

```
In [30]: var('x t')
curves = [x^3 - x, exp(sqrt(x)), log(x + 1)]
for f in curves:
    p_y = revolution_plot3d(f, (x, 0, 2), (t, 0, 2*pi), axis=(0, 0), color="red")
    p_x = revolution_plot3d(f, (x, 0, 2), (t, 0, 2*pi), axis=(1, 0), color="brown")
    show(p_y)
    show(p_x)
```







In []: